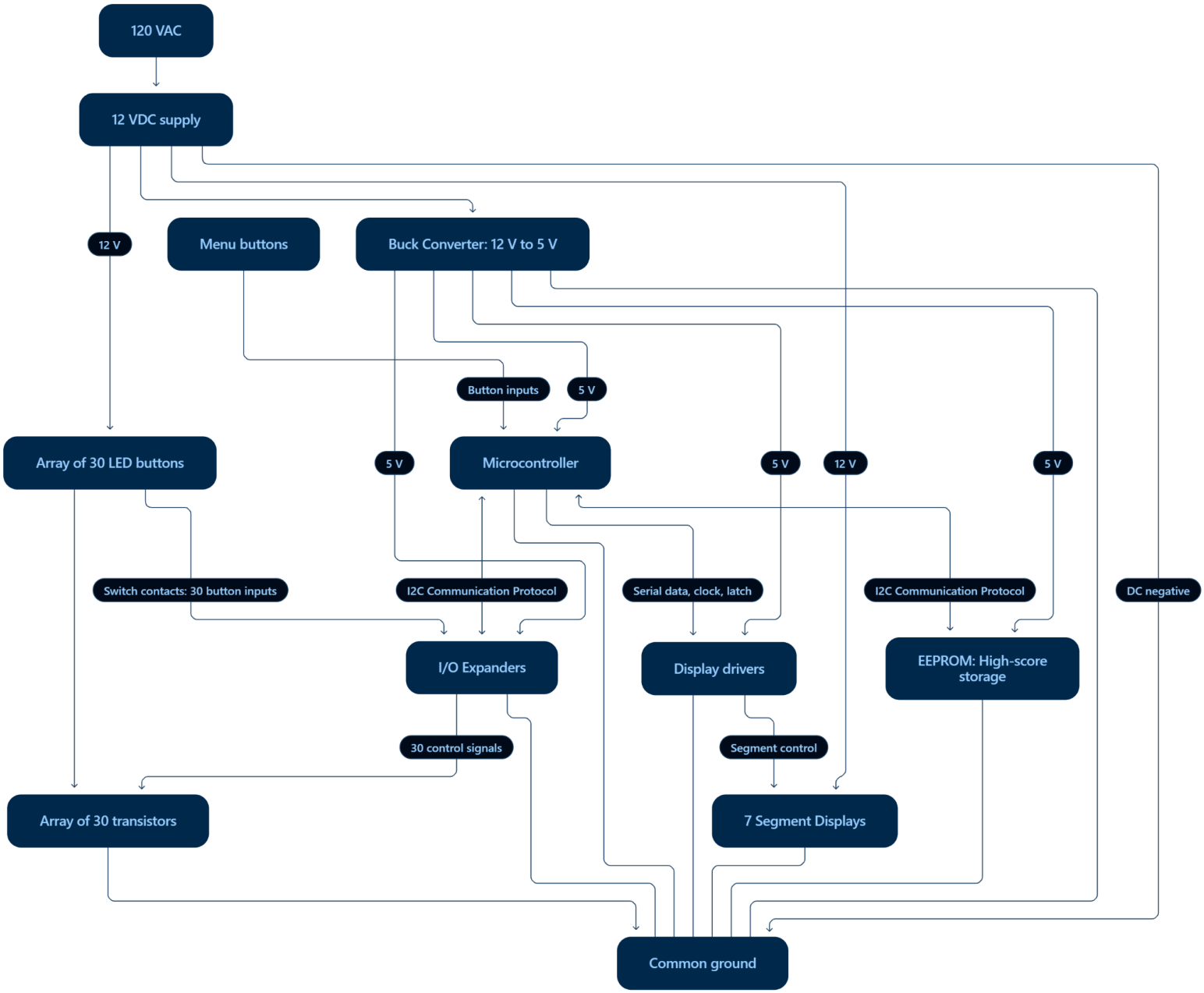
Whack-a-Volt/Whamperage Project Description

Speed of Light Game

Inspired by the arcade game Speed of Light, this project will feature a 6 × 5 array of 30 large illuminated pushbuttons with gameplay similar to whack-a-mole. During a timed round, such as 30 seconds, buttons will illuminate randomly, and players will earn points by pressing them. Faster reactions will allow players to score more points before time runs out. A three-digit seven-segment scoreboard will show the current score during play and the high score between rounds.

The game will include menu buttons for starting and restarting, with additional controls as needed. It will be housed in a lightweight wooden arcade-style cabinet and powered from a standard wall outlet, with an optional battery system for cordless play. The block diagram below illustrates the connections between the game's main components.



If more time allows, here are some additional features we can incorporate:

- Two player mode
- Different play modes (ex: easy, medium, hard, memory mode)
- Speaker and amplifier
- Countdown timer display

Skills this game will incorporate:

- Microcontrollers and Embedded Systems
- Coding and Debugging: RTOS, interrupts, polling, debouncing, state machines, timing and latency
- Communication protocols: I2C
    - IC Devices (with decoupling capacitors): I/O Expanders, Display drivers, EEPROM
- Power electronics: 120VAC → 12VDC → 5VDC
    - Protection: Power switch, fuse
    - Amperage and power calculations
- Transistors: Logic driven MOSFETS
    - Pull up/down resistors
- Git
- Instrumentation: Multimeters, Oscilloscopes
- PCB Design
    - BOM, part compatibility, datasheets
- Soldering: SMD, through hole
- Wire stripping, crimping, connecting, terminal blocks

Links to some parts:

- Large LED push buttons
- I/O Expanders
- MOSFETs
- Small push buttons